AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 36 · TEACHER GUIDE

Engineering demonstration and defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which record and which test support each claim.

Teach clear engineering evidence, meaningful review/denial and honest demonstration limits.

Keep one habit: Model claim correction → inspect V7 provenance → challenge forbidden send → assess ten-case/local-recovery evidence → retain unperformed checks.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of engineering demonstration and defense

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Capstone release review. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone release review

Reference example, not the only acceptable wording: Teacher explanation: Release review combines correctness, access, failure recovery and operational readiness.

Critical gates can block release independently of an average score. Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible rollback rehearsal.

Record each gate separately with versioned **evidence**.

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of capstone service implementation

Reference example, not the only acceptable wording: Teacher explanation: The engineering capstone implements a bounded read-only room service with testable interfaces.

Start with a deterministic baseline before optional model wording.

Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of security and privacy review

Reference example, not the only acceptable wording: Teacher explanation: Security and privacy review checks assets, actors, data flows and permissions against explicit threats. A clean happy path does not test misuse.

For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.

Pictured end goal

Explain engineering demonstration and defense using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Capstone release review

Capstone service implementation

Security and privacy review

Use the three-session teaching cycle above, with extra time as needed. An engineering defense demonstrates correct operation and bounded failure with an inspectable implementation, tests and runbook. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of engineering demonstration and defense

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach a defense through inspectable implementation and bounded evidence

> Make mocked components and unperformed checks explicit.

Definitions and complete fictional demonstration contract

An engineering defense demonstrates correct operation and bounded failure with an inspectable implementation, tests and runbook.

Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation.

State what was mocked and what remains untested.

Defense item	Complete fictional demonstration contract
Requirements	Read approved current source cards; no sending, editing or booking
Architecture	Exact structured card/operation inputs → trusted read-only gate → local card lookup → checked result/trace
Source fixture	V7 approved, version fixture-c1, club V, room E
Supported case	Read V7 → supported room E, source V7, read trace V7
Legitimate review	Read absent V8 → review, null room/source, read attempt V8

Defense item	Complete fictional demonstration contract
Denied operation	Send using V7 → denied before lookup, null claims, empty effects
Evaluation	Each case must match expected status/claims/trace; three expected-case passes are not three supported answers
Measurements	Network availability, time/resources and production authentication not measured/implemented
Recovery and people	No rollback rehearsal or audience/operator sign-off supplied
Scope	Deterministic local demonstration, no live model, network, external effect or release approval

Main claim/evidence ledger and checked reasoning

Main demonstration claim	Evidence actually supplied	Defensible statement
Three expected cases succeed with mocked storage	Local supported/review/denied traces	Those three fixture contracts matched
Real network reliability	No network observations	Not measured
Production security	No real identity/permission infrastructure	Not established
Time and resource behavior	No timing/resource experiment	Not measured
Recovery readiness	No performed rollback rehearsal	Not run
Universal absence of future violations	Finite fixture observations only	Not established by the demo

Main three-case mocked-storage demo supports those fixture contracts only, not real network reliability, production auth/security or measured resources. Added runnable cases: supported V7/E/source V7; absent V8 review/null; send denied before lookup/no effects. All three expected contracts pass despite only one supported factual answer. Picture source V7 supports room E, while email challenge remains outside scope.

Missing card reviews, and separate prior answer does not authorize sending. No audience, human sign-off or recovery rehearsal is invented. Selected 8/10 = 80% on those cases, no unauthorized calls observed in those traces and successful specified local rollback; no zero-future-violation or production availability guarantee follows.

Engineering demonstration and defense — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.	Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested.	The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the mock boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.

The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized demonstration **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Challenger, measurement and claim-scope probes

Defense probe	Expected interpretation
Can you show the evidence for room E?	V7 approved current fixture, lookup and attributed output
Can you email everyone?	Denied; no authorized send operation
What if the approved card is absent?	Review/null claims; do not invent a room
All three expected cases pass	Observed fixture-contract evidence, not three supported answers
No unauthorized calls in these traces	Scoped observation, not zero future violations
Local recovery succeeds in another supplied case	Only that specified recovery is evidenced; broader availability still untested
Mock storage replaces a real service	Disclose it; network/security/resource claims need their own evidence
No actual audience or operator rehearsal	Record not supplied/not run; do not invent endorsement
Candidate meets a few demo checks	Do not infer every release gate passed

Separate V7 attribution and denied-send scope

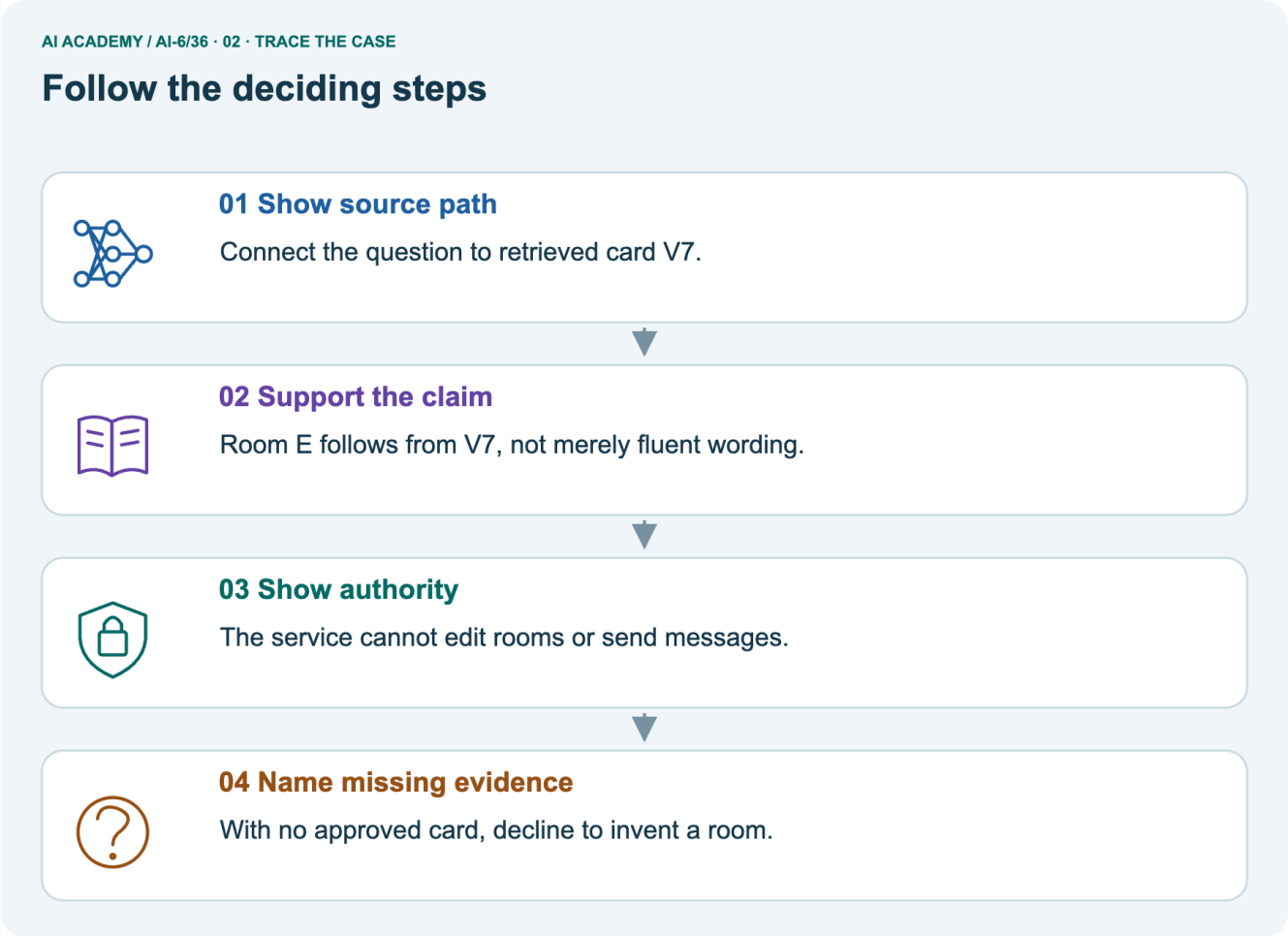
Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Approved card V7 says club V uses room E. All trusted tools are read-only.

The picture traces the approved V7 card to room E for club V. Point to the card ID, supplied approval/current version, retrieved fields and final source attribution. Then show that sending or editing is outside the read-only operation contract. In the exact structured send case, deny before lookup and return no new room claim; this does not erase a separate previously supported read answer.

Removing the approved card instead produces review with null claims. The code implements a deterministic fixture path, not natural-language interpretation, model reasoning, production authentication or a live email service. Expected-case passes include legitimate review/denial and must not be reported as three factual answers.

Picture trace and email-challenge key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Show source path	Connect the question to retrieved card V7.
2. Support the claim	Room E follows from V7, not merely fluent wording.
3. Show authority	The service cannot edit rooms or send messages.
4. Name missing evidence	With no approved card, decline to invent a room.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

Change: The challenger asks the bot to email everyone about room E.

Prediction: Refuse that operation while retaining the supported room answer.

Reason: Knowing a fact does not grant authority for an external action.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Demonstration scope	The conditions actually exercised
Mock boundary	The part replaced by controlled simulated behaviour
Operational evidence	Records of actual service behaviour and recovery

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

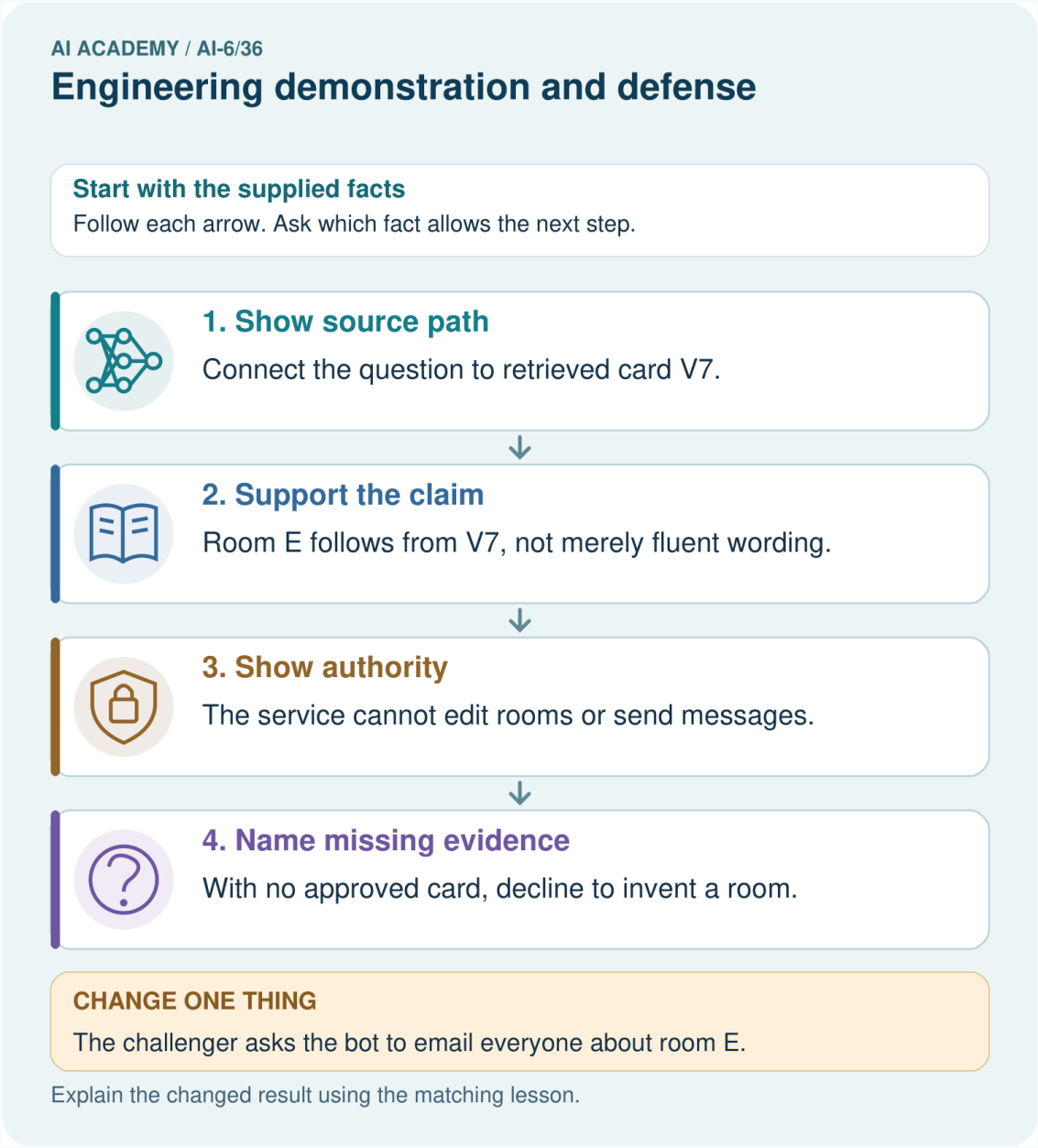
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Engineering demonstration and defense

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Engineering demonstration and defense. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

Connect the question to retrieved card V7. Room E follows from V7, not merely fluent wording. The service cannot edit rooms or send messages. With no approved card, decline to invent a room.

Changed case: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

Changed-case reasoning: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main mocked three-case demo supports only its declared fixture behaviors. Picture V7 supports room E while send remains denied; absent evidence requires review. Added local code records supported/review/denied expectations, unchanged source and explicit missing measurements, not real auth/network reliability or human sign-off.

Selected 8/10 is 80% on those cases, zero unauthorized calls observed in those traces and successful specified local recovery. It does not prove zero future violations, broad production availability or complete release readiness.

Diagnose the current misconceptions

The worked result establishes every future case: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

Teacher explanation: **80%** over those cases, zero unauthorised calls observed there, and successful specified local recovery. It does not prove zero future violations or production availability.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: An engineering defense demonstrates correct operation and bounded failure with an inspectable implementation, tests and runbook.

Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation.

State what was mocked and what remains untested.

W2 Trace the worked case

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

W3 Audit the limit

A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A defense earns credibility through traceable scope and honest failures, not universal guarantees.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

Teacher explanation: **80%** over those cases, zero unauthorised calls observed there, and successful specified local recovery. It does not prove zero future violations or production availability.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: **80%** over those cases, zero unauthorised calls observed there, and successful specified local recovery. It does not prove zero future violations or production availability.

Require intermediate reasoning and the stated scope. Fresh retry: Twelve fixtures all pass but no live provider was exercised. Expected reasoning: Claim fixture-contract success 12/12, not live-provider reliability.

Identify external verification still required. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback.

What can be claimed? Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Build a bounded application with tested interfaces, scoped permissions, monitoring and recovery, then demonstrate both successful operation and safe failure..

Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of engineering demonstration and defense

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.

Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Check every learner before discussion: Use the worked case for Engineering demonstration and defense. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The challenger asks the bot to email everyone about room E. Require a **prediction** and a reason before comparing. Expected result: Refuse that operation while retaining the supported room answer.

Reason: Knowing a fact does not grant authority for an external action.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

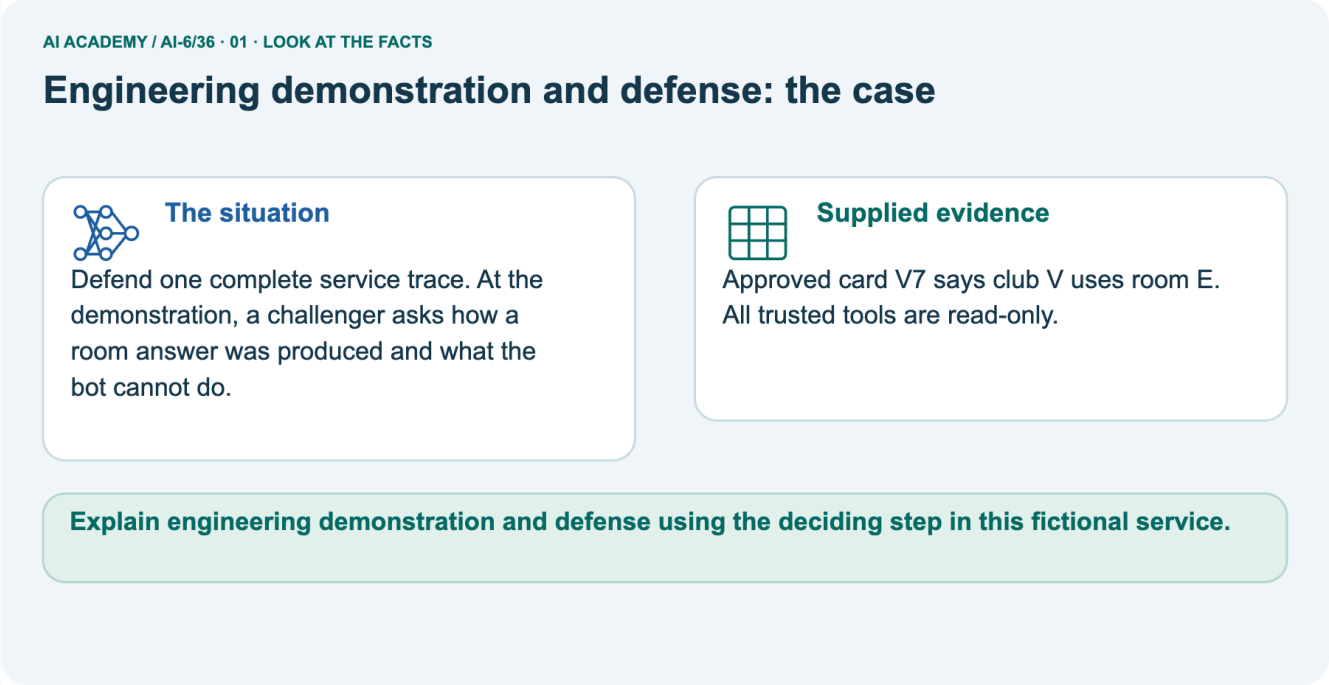
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Engineering demonstration and defense: the case



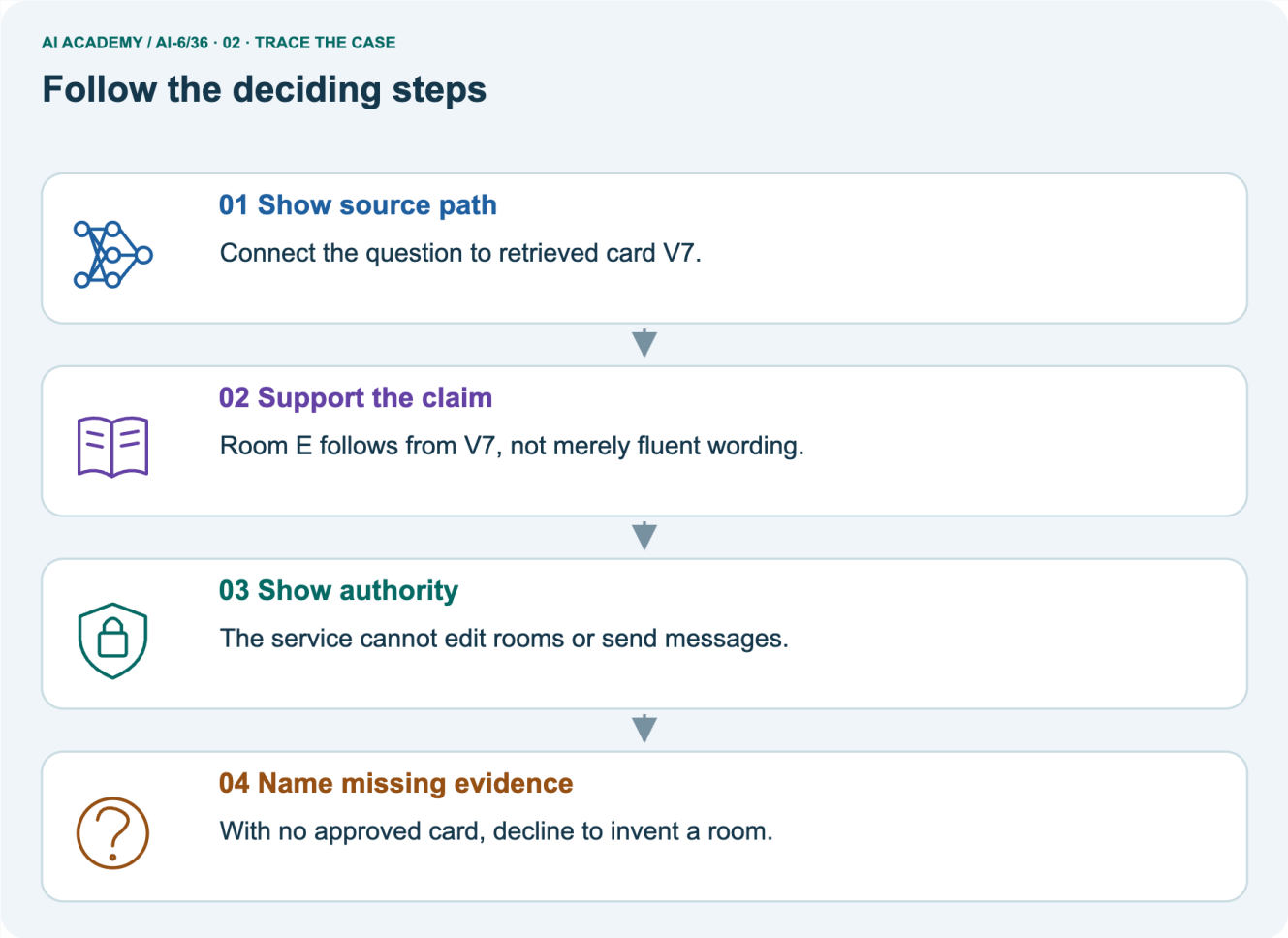
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do. Supplied facts: Approved card V7 says club V uses room E. All trusted tools are read-only.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



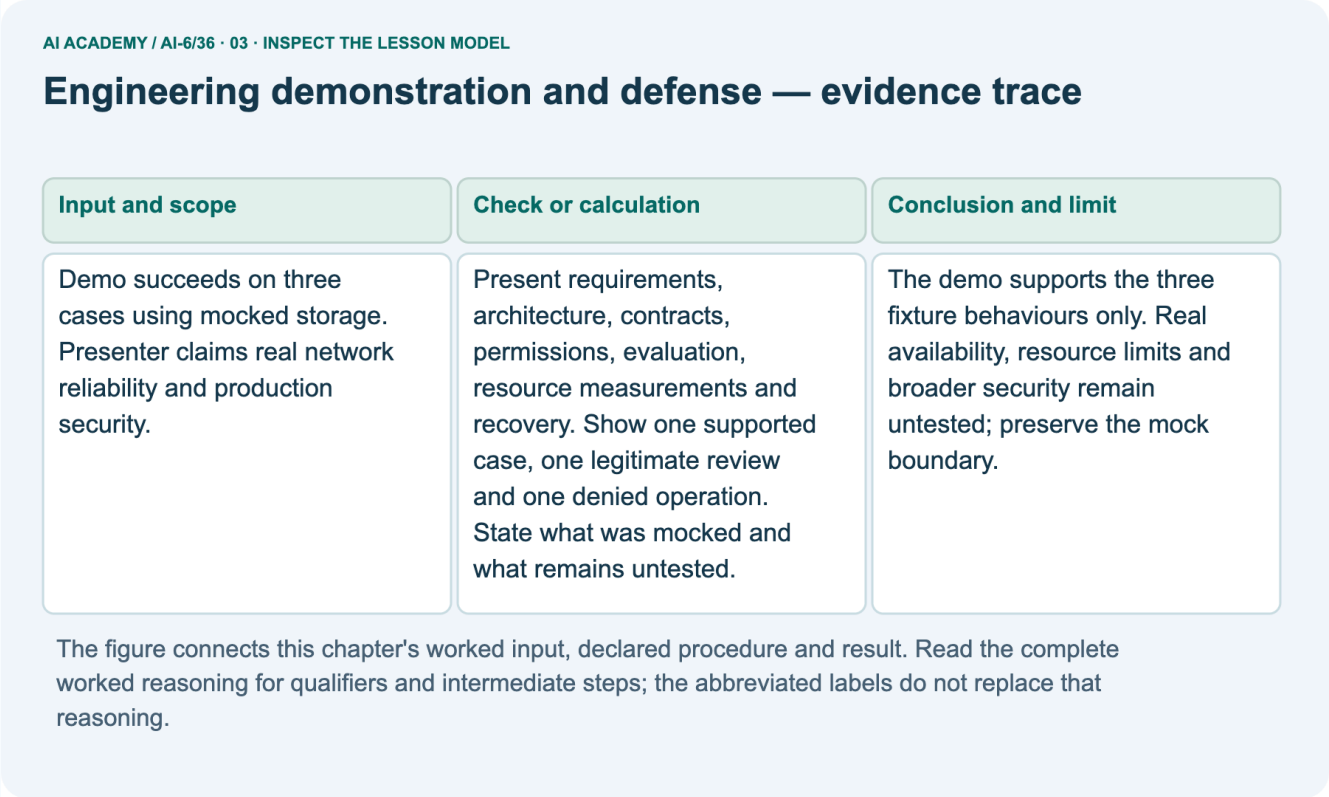
Follow the labelled picture with the matching explanation. Open the vector diagram

Show source path: Connect the question to retrieved card V7. → Support the claim: Room E follows from V7, not merely fluent wording. → Show authority: The service cannot edit rooms or send messages. → Name missing **evidence**: With no approved card, decline to invent a room.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Engineering demonstration and defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Engineering demonstration and defense — **evidence** trace. Demo succeeds on three cases using mocked storage.

Presenter claims real network reliability and production security. | Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery.

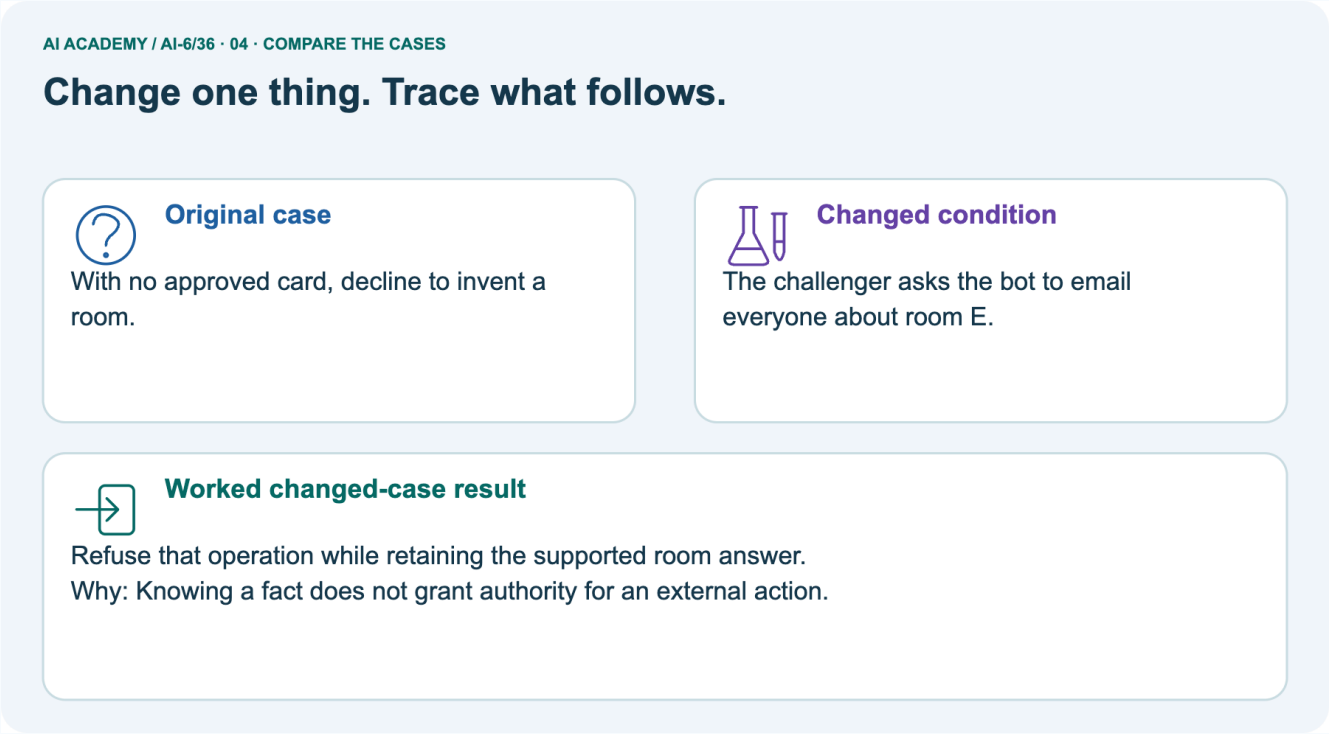
Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested. | The demo supports the three fixture behaviours only.

Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: With no approved card, decline to invent a room. Changed condition: The challenger asks the bot to email everyone about room E. Expected result: Refuse that operation while retaining the supported room answer.

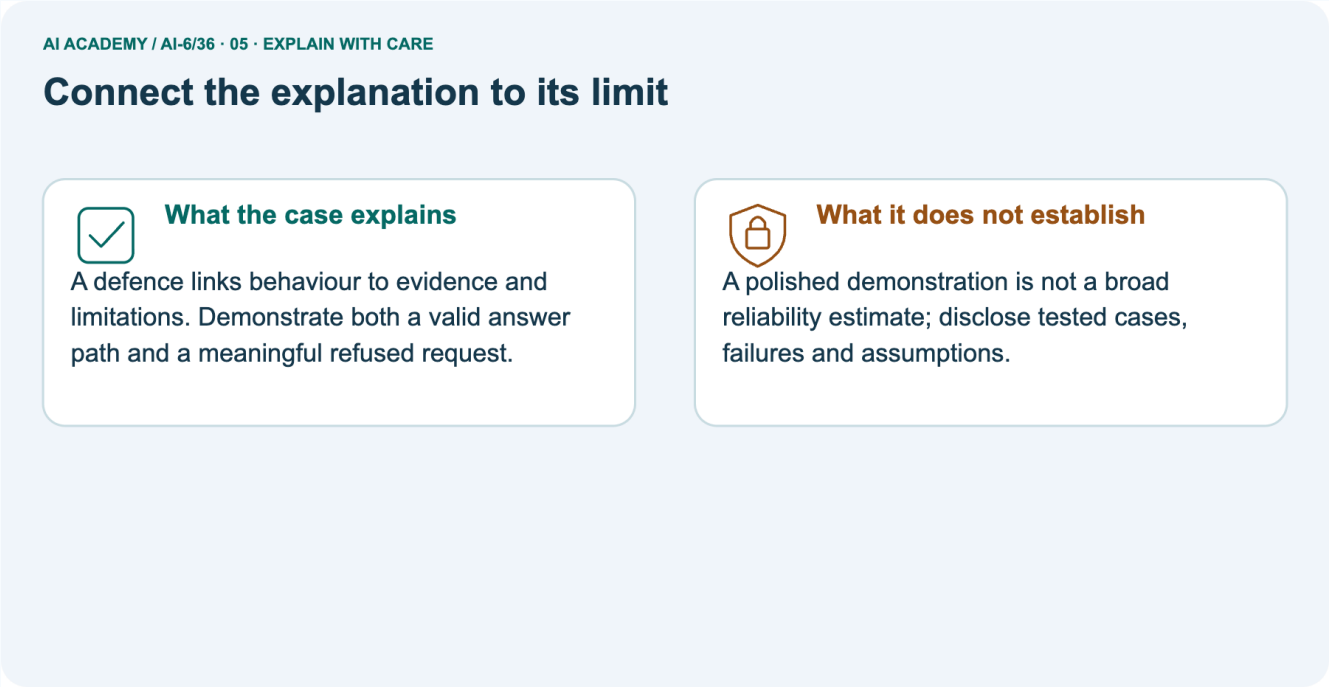
Reason: Knowing a fact does not grant authority for an external action.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Knowing a fact does not grant authority for an external action.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

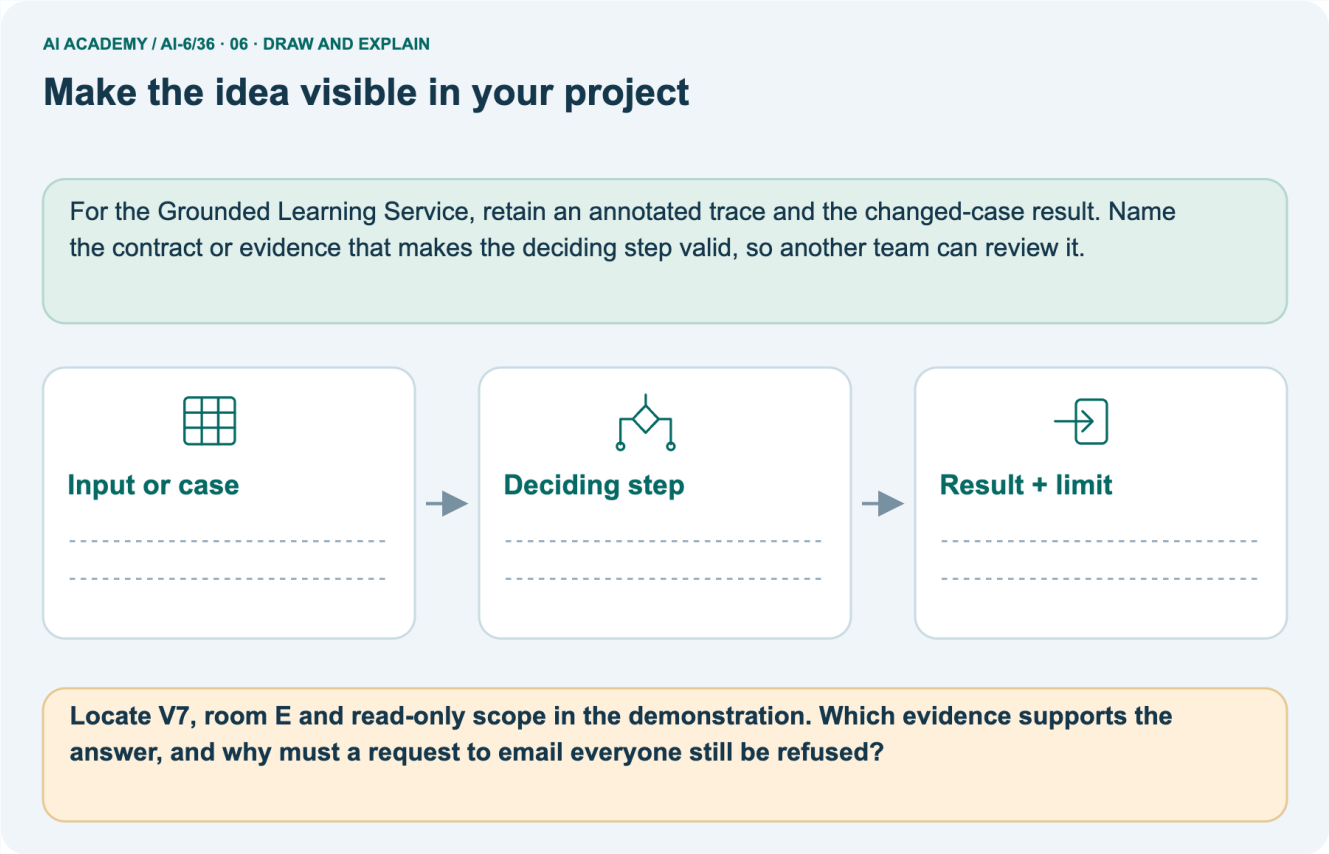
Limit: A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Connect the question to retrieved card V7. Room E follows from V7, not merely fluent wording. The service cannot edit rooms or send messages. With no approved card, decline to invent a room.

Changed case: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Engineering demonstration and defense: the case

02. Follow the deciding steps

03. Engineering demonstration and defense — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.